



BA UNIT2 - notes

Big data analytics (Smt. Kashibai Navale College Engineering)



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UNIT II BUSINESS INTELLIGENCE

Data Warehouses and Data Mart - Knowledge Management – Types of Decisions - Decision Making Process - Decision Support Systems – Business Intelligence – OLAP – Analytic functions



DATA WAREHOUSES AND DATA MART**INTRODUCTION**

Business intelligence, as we know it today, would not be possible without the data warehouse. At its core, business intelligence is the ability to answer complex questions about your data and use those answers to make informed business decisions. In order to do this well, you need a data warehouse, which not only provides a safe way to centralize and store all your data but also a method to quickly find the answers you need, when you need them. And that's a pretty important role. By 2025, it's estimated humanity will have produced a total of 175 zettabytes of data. For context, that's 175,000,000,000 terabytes.

Where does all of this information go? Well, most of it goes in the data warehouses.

Companies use data warehouses to manage transactions, understand their data, and keep it all organized. In short, data warehouses make large amounts of information more usable for organizations of all sizes and types. This has made them a linchpin of data pipelines and business intelligence systems the world over. And understanding how data warehouses work can help you fulfill the full potential of business intelligence (it's not as complex as it may seem).

WHAT IS A DATABASE?

A database is a storage location of related data used to capture a specific situation. One example of a database is a point-of-sale (POS) database. The POS database will capture and store all the relevant data surrounding a retail store's transactions.

Databases have a variety of flavors: structured, relational, relational database management systems (RDBMS), or unstructured data structures (known as 'NoSQL'). New data coming into the database is processed, organized, managed, updated and then stored in tables. Databases are single-purpose repositories of raw transactional data. Because a database is closely tied with transactions, a database performs online transactional processing (OLTP).

Main Characteristics of Databases

- Structured according to company operations and applications
- Rigid rules set around data storage/organization (RDBMS-specific)
- Flexible data storage (NoSQL-specific)
- Single-purpose in its nature: handles one process (e.g., POS)
- Utilized for online transaction processing (OLTP)
- Data recording capabilities, capturing transactions as they occur and housing those transactions

WHAT IS A DATA WAREHOUSE?

A data warehouse is a data management system that stores large amounts of data for later use in processing and analysis. You can think of it as a large warehouse where trucks (i.e., source data) unload their data. That data is then sorted into rows and rows of well-organized shelves that make it easy to find exactly what you're looking for later. The biggest innovation data warehouses introduced at their inception, according to DW 2.0: The Architecture for the Next Generation of Data Warehousing, was the ability to store "integrated granular historical data."

Breaking that down into human terms, this means data warehouses excel at storing data that's:

- Integrated: They combine data from many databases and data sources.
- Granular: The data they house is highly detailed and can be used in many different ways.
- Historical: They can host a continuous record of data over years and years.

You can store this data in three different ways: on-premise data warehouses, cloud data warehouses, and hybrid data warehouses.

On-premise data warehouses run on physical servers that your company owns and manages. Cloud data warehouses are fully online, and you pay for space on servers that another company manages, like Amazon Redshift. Hybrid data warehouses are a mix of both on-premise and cloud, and companies making the transition to the cloud over a period of time use this option.

With all the data stored in one place, data warehouses use a specific approach to process data called online analytical processing (OLAP), which is specifically designed for complex queries. One way to think about it is that when you go to your data warehouse to ask a question about the relationship between one set of data and another, OLAP is a way of organizing and moving among the rows and rows of shelves to quickly find that information.

This is great for business intelligence because the questions you ask about your data in order to make decisions are rarely simple. Because data warehouses use OLAP, they make finding answers to these complex questions very efficient. As a result, they've become a foundation for many successful business intelligence systems.

What Is the Role of Data Warehousing in Business Intelligence?

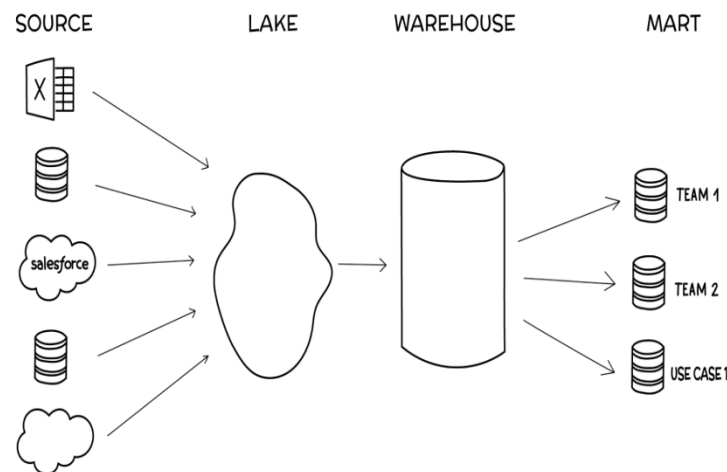
In business intelligence, data warehouses serve as the backbone of data storage. Business intelligence relies on complex queries and comparing multiple sets of data to inform everything from everyday decisions to organization-wide shifts in focus.

To facilitate this, business intelligence is comprised of three overarching activities: data wrangling, data storage, and data analysis. Data wrangling is usually facilitated by extract, transform, load (ETL) technologies, which we'll explain in detail below, and data analysis is done using business intelligence tools, like Chartio.

The glue holding this process together is data warehouses, which serve as the facilitator of data storage using OLAP. They integrate, summarize, and transform data, making it easier to analyze. Even though data warehouses serve as the backbone of data storage, they're not the only technology involved in data storage. Many companies go through a data storage hierarchy before reaching the point where they absolutely need a data warehouse.

When Should I Use a Data Warehouse for Business Intelligence?

There are generally four stages of data sophistication: source data, data lakes, data warehouses, and data marts. Knowing when to invest in a data warehouse requires you to know each stage, but at the end of the day, the data warehouse stage is what unlocks the true power of your data.



SOURCE DATA

Source data is any individual set of data like databases, Excel spreadsheets, individual application reports, etc. It's structured (i.e., organized) yet siloed data that works fine alone but does not provide a larger picture of your organization's data as a whole.

DATA LAKE

For teams who have graduated to a need to centralize their source data into one place, a data lake is increasingly becoming the next step. A data lake serves as a central repository for all raw, unstructured (i.e., not organized) data.

If a data warehouse is like backing up a truck and unloading the data in an orderly fashion into a well-organized shelving system, data lakes are like backing the truck up and dumping all the data into, well, a lake. James Dixon, who coined the term "data lake," describes it as the natural raw state of data that, for people with the diving skills, serves as a frontier to explore.

The drawback of a data lake is that the data is not ready for analysis. It's not well-organized, there may be duplicates, and in order to make sense of it, you'll need to tell your diver exactly what you're looking for. Even then, the diver might not find exactly what you need after all that effort.

DATA WAREHOUSE

Like a data lake, a data warehouse centralizes your data, but as we've established, it's well-organized and set up for efficient analysis. It's a single source of truth for all data that's easier to understand and navigate.

Data warehouses can hook right up to source data, but nowadays, we're seeing more and more companies use their data warehouse as a layer on top of their data lake. Following Dixon's comparison, if a data lake is the water/data in its natural, unorganized state, a data warehouse is where you treat it and make it ready for consumption.

Building a Data Warehouse: The Basics

Business Intelligence has advanced quickly and dramatically in recent years, and many people are taking advantage of it. To be the most successful and efficient with this newfound Business Intelligence (BI) power, it's essential to be able to analyze and harness ALL of your data. Enter the data warehouse.

Simply put, a data warehouse is a large store of data that's collected from multiple different sources within a business. A data warehouse is used as storage for data analytic work (OLAP systems), leaving the transactional database (OLTP systems) free to focus on transactions. With a significant amount of data kept in one place, it's now easier for businesses to analyze and make better-informed decisions.

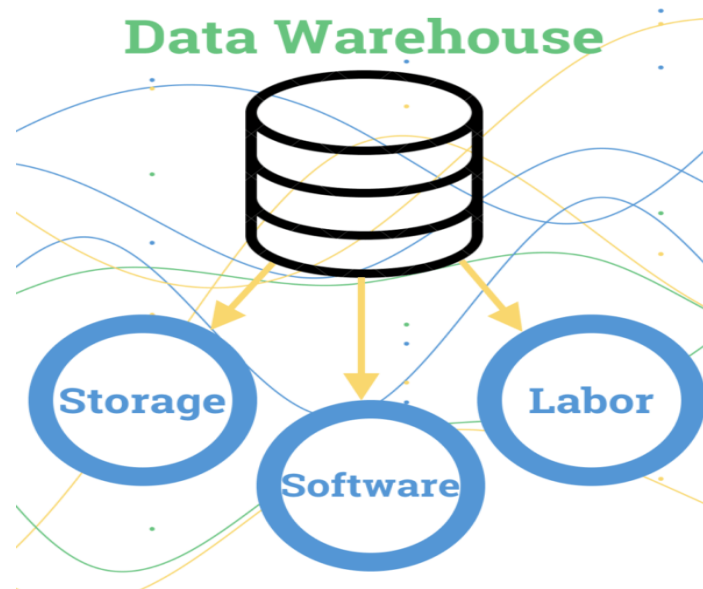
Why Should I Build a Data Warehouse?

While having all of your data gathered in one place is arguably the biggest benefit of having a data warehouse, it is certainly not the only one. Here, we've listed some of the other benefits of having a data warehouse:

- **Save Time** – Business users can quickly access data from multiple sources within a data warehouse, meaning that time won't be wasted on retrieving data from multiple sources.
- **Boost Confidence** – Having data transferred automatically to your data warehouse by a structured system, as opposed to being transferred by human labor, gives you more confidence that your data is clean, current and complete.
- **Increase Insight** – Data warehouses structure your data so it's easily analyzable.
- **Improve Security** – Managing who has access to your data is much easier when there's a centralized connection point. Data warehouses make security completely customizable, so you're able to give access to whoever you'd like and lock down all of your other systems.

When using a data warehouse to its full potential, analyzing data becomes convenient and answering important questions about your business becomes simple. Your data is organized and available so you can get your answers quickly and securely.

STRUCTURE OF A DATA WAREHOUSE



Regardless of the specific approach, you take to building a data warehouse, there are three components that should make up your basic structure: A storage mechanism, operational software, and human resources.

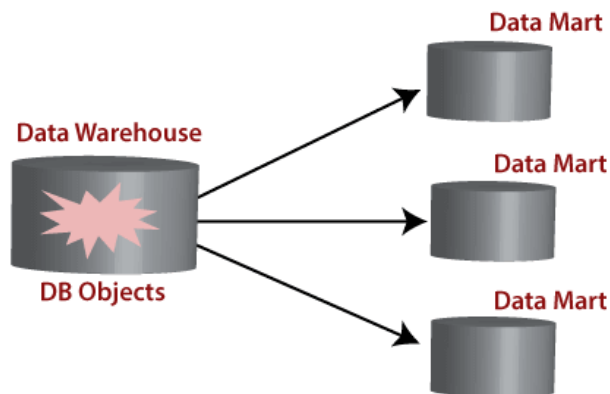
- **Storage** – This part of the structure is the main foundation — it's where your warehouse will live. There are two main options when it comes to storage, an in-house server (Oracle, Microsoft SQL Server) or on the cloud (Amazon S3, Microsoft Azure). An in-house server is internal hardware that's set up within your office, and the cloud is a digital storage solution based on external servers. Either is a feasible option when it comes to storage and all depends on your needs.
- **Software** – This is the operational part of the data warehouse structure. It's often broken down into two categories — centralization software and visualization software. Centralization software is needed to collect and maintain the data that comes from all of your separate databases. Visualization software is needed to take the data and present it in a visual form to aid in analyzation. Some centralization software includes visualization software as part of its package, but it is highly recommended that you have both types of software regardless.
- **Labor** – This is the management aspect of the data warehouse, something that's absolutely essential in having a working solution. To keep your warehouse functional, it might be necessary to hire new positions within your business. Hiring well-skilled professionals is crucial, as running a data warehouse requires a lot of knowledge. However, if you choose to have a cloud-based warehouse, it might not be necessary to

have as many human resources. The cloud is managed by third-party vendors, so it's their responsibility to do routine maintenance on hardware and servers

WHAT IS DATA MART?

A **Data Mart** is a subset of a directorial information store, generally oriented to a specific purpose or primary data subject which may be distributed to provide business needs. Data Marts are analytical record stores designed to focus on particular business functions for a specific community within an organization. Data marts are derived from subsets of data in a data warehouse, though in the bottom-up data warehouse design methodology, the data warehouse is created from the union of organizational data marts.

The fundamental use of a data mart is **Business Intelligence (BI)** applications. **BI** is used to gather, store, access, and analyze record. It can be used by smaller businesses to utilize the data they have accumulated since it is less expensive than implementing a data warehouse.



Reasons for creating a data mart

- Creates collective data by a group of users
- Easy access to frequently needed data
- Ease of creation
- Improves end-user response time
- Lower cost than implementing a complete data warehouses
- Potential clients are more clearly defined than in a comprehensive data warehouse
- It contains only essential business data and is less cluttered.

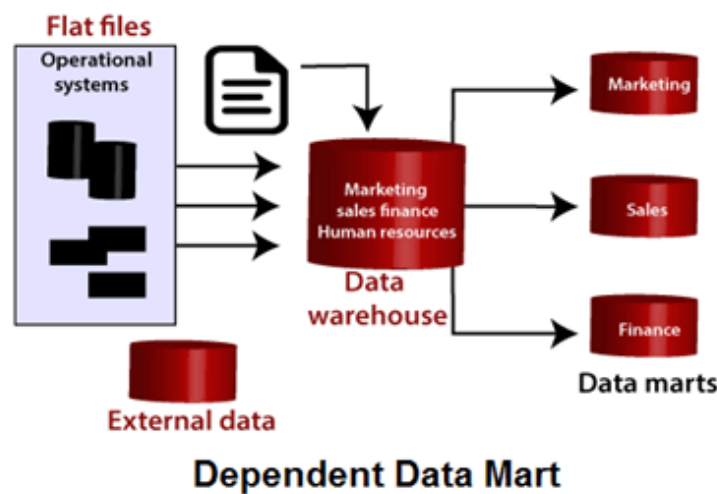
Types of Data Marts

There are mainly two approaches to designing data marts. These approaches are

- Dependent Data Marts
- Independent Data Marts

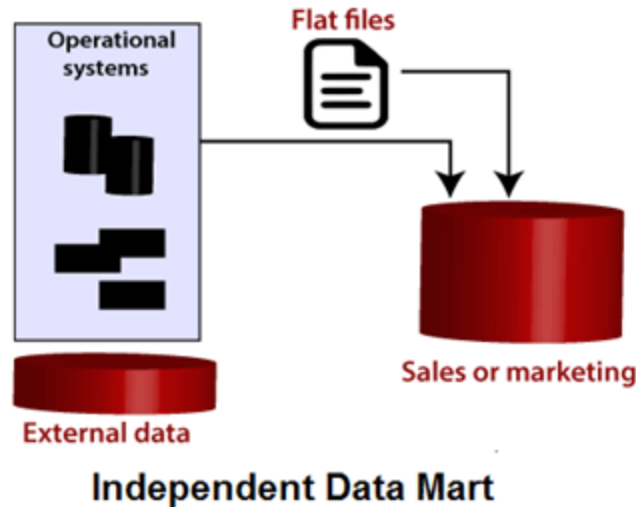
Dependent Data Marts

A dependent data marts is a logical subset of a physical subset of a higher data warehouse. According to this technique, the data marts are treated as the subsets of a data warehouse. In this technique, firstly a data warehouse is created from which further various data marts can be created. These data mart are dependent on the data warehouse and extract the essential record from it. In this technique, as the data warehouse creates the data mart; therefore, there is no need for data mart integration. It is also known as a **top-down approach**.



Independent Data Marts

The second approach is Independent data marts (IDM). Here, firstly independent data marts are created, and then a data warehouse is designed using these independent multiple data marts. In this approach, as all the data marts are designed independently; therefore, the integration of data marts is required. It is also termed as a **bottom-up approach** as the data marts are integrated to develop a data warehouse.



Other than these two categories, one more type exists that is called "**Hybrid Data Marts.**"

Hybrid Data Marts

It allows us to combine input from sources other than a data warehouse. This could be helpful for many situations; especially when Adhoc integrations are needed, such as after a new group or product is added to the organizations.

Steps in Implementing a Data Mart

The significant steps in implementing a data mart are to design the schema, construct the physical storage, populate the data mart with data from source systems, access it to make informed decisions and manage it over time. So, the steps are:

Designing

The design step is the first in the data mart process. This phase covers all of the functions from initiating the request for a data mart through gathering data about the requirements and developing the logical and physical design of the data mart.

It involves the following tasks:

1. Gathering the business and technical requirements
2. Identifying data sources
3. Selecting the appropriate subset of data
4. Designing the logical and physical architecture of the data mart.

Constructing

This step contains creating the physical database and logical structures associated with the data mart to provide fast and efficient access to the data.

It involves the following tasks:

1. Creating the physical database and logical structures such as table spaces associated with the data mart.
2. Creating the schema objects such as tables and indexes describe in the design step.
3. Determining how best to set up the tables and access structures.

Populating

This step includes all of the tasks related to the getting data from the source, cleaning it up, modifying it to the right format and level of detail, and moving it into the data mart.

It involves the following tasks:

1. Mapping data sources to target data sources
2. Extracting data
3. Cleansing and transforming the information.
4. Loading data into the data mart
5. Creating and storing metadata

Accessing

This step involves putting the data to use: querying the data, analyzing it, creating reports, charts and graphs and publishing them.

It involves the following tasks:

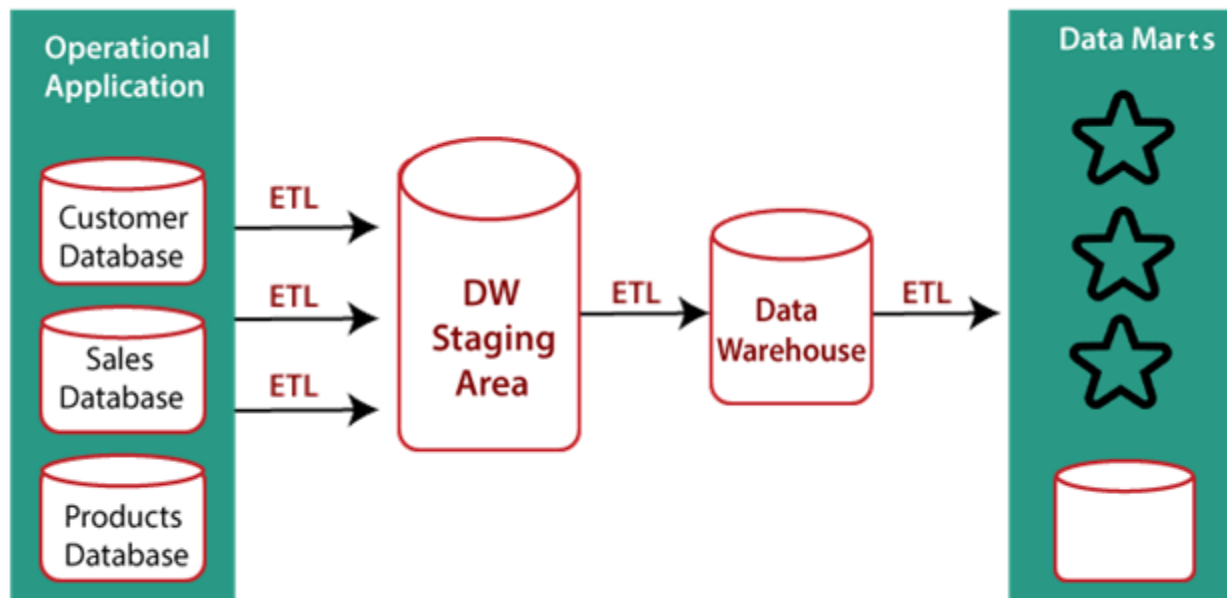
1. Set up an intermediate layer (Meta Layer) for the front-end tool to use. This layer translates database operations and objects names into business conditions so that the end-clients can interact with the data mart using words which relates to the business functions.
2. Set up and manage database architectures like summarized tables which help queries agree through the front-end tools execute rapidly and efficiently.

Managing

This step contains managing the data mart over its lifetime. In this step, management functions are performed as:

1. Providing secure access to the data.
2. Managing the growth of the data.
3. Optimizing the system for better performance.
4. Ensuring the availability of data event with system failures.

Difference between Data Warehouse and Data Mart



Data Warehouse	Data Mart
A Data Warehouse is a vast repository of information collected from various organizations or departments within a corporation.	A data mart is an only subtype of a Data Warehouses. It is architecture to meet the requirement of a specific user group.
It may hold multiple subject areas.	It holds only one subject area. For example, Finance or Sales.

It holds very detailed information.	It may hold more summarized data.
Works to integrate all data sources	It concentrates on integrating data from a given subject area or set of source systems.
In data warehousing, Fact constellation is used.	In Data Mart, Star Schema and Snowflake Schema are used.
It is a Centralized System. It is a Decentralized System.	
Data Warehousing is the data-oriented.	Data Marts is a project-oriented.

CONCLUSION

A data warehouse is a great solution to centralizing and easily analyzing your business's data. It increases data availability, boosts efficiency in analytical activity, improves the quality of information needed for reporting, and makes working with data secure. The structure of a data warehouse is basic, consisting of a storage system, two types of software, and a few employees to make it all work.

KNOWLEDGE MANAGEMENT

WHAT IS KNOWLEDGE MANAGEMENT?

Knowledge management (KM) is the process of identifying, organizing, storing and disseminating information within an organization. When knowledge is not easily accessible within an organization, it can be incredibly costly to a business as valuable time is spent seeking out relevant information versus completing outcome-focused tasks.

A knowledge management system (KMS) harnesses the collective knowledge of the organization, leading to better operational efficiencies. These systems are supported by the use of a knowledge base. They are usually critical to successful knowledge management, providing a centralized place to store information and access it readily.

Companies with a knowledge management strategy achieve business outcomes more quickly as increased organizational learning and collaboration among team members facilitates faster decision-making across the business. It also streamlines more organizational processes, such as training and on-boarding, leading to reports of higher employee satisfaction and retention.

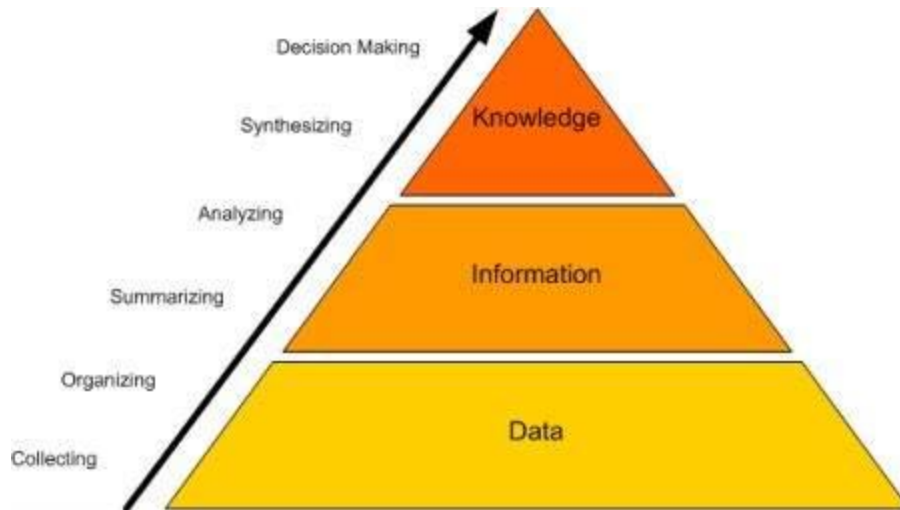
TYPES OF KNOWLEDGE

The definition of knowledge management also includes three types of knowledge—tacit, implicit, and explicit knowledge. These types of knowledge are largely distinguished by the codification of the information.

- **Tacit knowledge:** This type of knowledge is typically acquired through experience, and it is intuitively understood. As a result, it is challenging to articulate and codify, making it difficult to transfer this information to other individuals. Examples of tacit knowledge can include language, facial recognition, or leadership skills.
- **Implicit knowledge:** While some literature equivocates implicit knowledge to tacit knowledge, some academics break out this type separately, expressing that the definition of tacit knowledge is more nuanced. While tacit knowledge is difficult to codify, implicit knowledge does not necessarily have this problem. Instead, implicit information has yet to be documented. It tends to exist within processes, and it can be referred to as “know-how” knowledge.
- **Explicit knowledge:** Explicit knowledge is captured within various document types such as manuals, reports, and guides, allowing organizations to easily share knowledge across teams. This type of knowledge is perhaps the most well-known and examples of it include knowledge assets such as databases, white papers, and case studies. This form of knowledge is important to retain intellectual capital within an organization as well as facilitate successful knowledge transfer to new employees.

THE KNOWLEDGE MANAGEMENT PROCESS

The process of knowledge management is universal for any enterprise. Sometimes, the resources used, such as tools and techniques, can be unique to the organizational environment. The Knowledge Management process has six basic steps assisted by different tools and techniques. When these steps are followed sequentially, the data transforms into knowledge.



Step 1: Collecting

This is the most important step of the knowledge management process. If you collect the incorrect or irrelevant data, the resulting knowledge may not be the most accurate. Therefore, the decisions made based on such knowledge could be inaccurate as well.

There are many methods and tools used for data collection. First of all, data collection should be a procedure in knowledge management process. These procedures should be properly documented and followed by people involved in data collection process.

The data collection procedure defines certain data collection points. Some points may be the summary of certain routine reports. As an example, monthly sales report and daily attendance reports may be two good resources for data collection.

With data collection points, the data extraction techniques and tools are also defined. As an example, the sales report may be a paper-based report where a data entry operator needs to feed the data manually to a database whereas, the daily attendance report may be an online report where it is directly stored in the database.

In addition to data collecting points and extraction mechanism, data storage is also defined in this step. Most of the organizations now use a software database application for this purpose.

Step 2: Organizing

The data collected need to be organized. This organization usually happens based on certain rules. These rules are defined by the organization.

As an example, all sales-related data can be filed together and all staff-related data could be stored in the same database table. This type of organization helps to maintain data accurately within a database.

If there is much data in the database, techniques such as 'normalization' can be used for organizing and reducing the duplication.

This way, data is logically arranged and related to one another for easy retrieval. When data passes step 2, it becomes information.

Step 3: Summarizing

In this step, the information is summarized in order to take the essence of it. The lengthy information is presented in tabular or graphical format and stored appropriately.

For summarizing, there are many tools that can be used such as software packages, charts (Pareto, cause-and-effect), and different techniques.

Step 4: Analyzing

At this stage, the information is analyzed in order to find the relationships, redundancies and patterns.

An expert or an expert team should be assigned for this purpose as the experience of the person/team plays a vital role. Usually, there are reports created after analysis of information.

Step 5: Synthesizing

At this point, information becomes knowledge. The results of analysis (usually the reports) are combined together to derive various concepts and artefacts.

A pattern or behavior of one entity can be applied to explain another, and collectively, the organization will have a set of knowledge elements that can be used across the organization.

This knowledge is then stored in the organizational *knowledge base* for further use.

Usually, the knowledge base is a software implementation that can be accessed from anywhere through the Internet.

You can also buy such knowledge base software or download an open-source implementation of the same for free.

Step 6: Decision Making

At this stage, the knowledge is used for decision making. As an example, when estimating a specific type of a project or a task, the knowledge related to previous estimates can be used.

This accelerates the estimation process and adds high accuracy. This is how the organizational knowledge management adds value and saves money in the long run.

Knowledge management tools

There are number of tools that organizations utilize to reap the benefits of knowledge management. Examples of knowledge management systems can include:

- **Document management systems** act as a centralized storage system for digital documents, such as PDFs, images, and word processing files. These systems enhance employee workflows by enabling easy retrieval of documents, such as lessons learned.

- **Content management systems (CMS)** are applications which manage web content where end users can edit and publish content. These are commonly confused with document management systems, but CMSs can support other media types, such as audio and video.
- **Intranets** are private networks that exist solely within an organization, which enable the sharing of enablement, tools, and processes within internal stakeholders. While they can be time-consuming and costly to maintain, they provide a number of groupware services, such as internal directories and search, which facilitate collaboration.
- **Wikis** can be a popular knowledge management tool given its ease of use. They make it easy to upload and edit information, but this ease can lead to concerns about misinformation as workers may update them with incorrect or outdated information.
- **Data warehouses** aggregate data from different sources into a single, central, consistent data store to support data analysis, data mining, artificial intelligence (AI), and machine learning. Data is extracted from these repositories so that companies can derive insights, empowering employees to make data-driven decisions.

STRATEGIES TO ACCELERATE KNOWLEDGE MANAGEMENT

While knowledge management solutions can be helpful in facilitating knowledge transfer across teams and individuals, they also depend on user adoption to generate positive outcomes. As a result, organizations should not minimize the value of human elements that enable success around knowledge management.

- **Organizational Culture:** Management practices will affect the type of organization that executives lead. Managers can build learning organizations by rewarding and encouraging knowledge sharing behaviors across their teams. This type of leadership sets the groundwork for teams to trust each other and communicate more openly to achieve business outcomes.
- **Communities of practice:** Centers of excellence in specific disciplines provide employees with a forum to ask questions, facilitating learning and knowledge transfer. In this way, organizations increase the number of subject matter experts in a given area of the company, reducing dependencies on specific individuals to execute certain tasks.

KNOWLEDGE MANAGEMENT USE CASES

Armed with the right tools and strategies, knowledge management practices have seen success in specific applications, such as:

- **Onboarding employees:** Knowledge management systems help to address the huge learning curve for new hires. Instead of overwhelming new hires with a ‘data dump’ in their first weeks, continually support them with knowledge tools that will give them useful information at any time.
- **Day-to-day employee tasks:** Enable every employee to have access to accurate answers and critical information. Access to highly relevant answers at the right time, for the right person, allows workforces to spend less time looking for information and more time on activities that drive business.

- **Self-serve customer service:** Customers repeatedly say they'd prefer to find an answer themselves, rather than pick up the phone to call support. When done well, a knowledge management system helps businesses decrease customer support costs and increase customer satisfaction.

BENEFITS OF KNOWLEDGE MANAGEMENT

Companies experience a number of benefits when they embrace knowledge management strategies. Some key advantages include:

- **Identification of skill gaps:** When teams create relevant documentation around implicit or tacit knowledge or consolidate explicit knowledge, it can highlight gaps in core competencies across teams. This provides valuable information to management to form new organizational structures or hire additional resources.
- **Make better informed decisions:** Knowledge management systems arm individuals and departments with knowledge. By improving accessibility to current and historical enterprise knowledge, your teams can upskill and make more information-driven decisions that support business goals.
- **Maintains enterprise knowledge:** If your most knowledgeable employees left tomorrow, what would your business do? Practicing internal knowledge management enables businesses to create an organizational memory. Knowledge held by your long-term employees and other experts, then make it accessible to your wider team.
- **Operational efficiencies:** Knowledge management systems create a go-to place that enable knowledge workers to find relevant information more quickly. This, in turn, reduces the amount of time on research, leading to faster decision-making and cost-savings through operational efficiencies. Increase productivity not only saves time, but also reduces costs.
- **Increased collaboration and communication:** Knowledge management systems and organizational cultures work together to build trust among team members. These information systems provide more transparency among workers, creating more understanding and alignment around common goals. Engaged leadership and open communication create an environment for teams to embrace innovation and feedback.
- **Data Security:** Knowledge management systems enable organizations to customize permission control, viewership control and the level of document-security to ensure that information is shared only in the correct channels or with selected individuals. Give your employees the autonomy access knowledge safely and with confidence.

CONCLUSION

Knowledge management is an essential practice for enterprise organizations. Organizational knowledge adds long-term benefits to the organization in terms of finances, culture and people. Therefore, all mature organizations should take necessary steps for knowledge management in order to enhance the business operations and organization's overall capability.

TYPES OF DECISIONS IN BUSINESS INTELLIGENCE**INTRODUCTION**

Modern cloud-native organizations have constantly growing streams of raw data flowing from every corner of the enterprise. Determining the impact this data has on business performance can be an overwhelming task requiring teams of analysts. That's where employing business intelligence (BI) can help. By presenting current and historical data within a business context, the data insights supplied by BI tools enable organizations to make smarter, more confident decisions that provide strategic direction for years to come.

Instead of relying on intuition and "gut feel," companies can use BI to find new ways to increase revenue, track performance, boost operational efficiency, identify market trends, expose problems, and much, much more. Before we dig deeper into the primary types of decisions in business intelligence, let's define what we mean by decision-making in a business context and understand how business intelligence factors into the process.

DECISION-MAKING DEFINED

Simply put, decision-making is the process of deciding something, especially with a group of people. From a business decision perspective, the aim is to achieve business objectives to satisfy stakeholder requirements, needs, and expectations.

For the decision to be effective, however, decision makers must forecast the outcome of each option and determine which is best for a particular situation. That makes decision support systems (DSS) like decision intelligence and business intelligence absolute essentials.

WHAT IS BUSINESS INTELLIGENCE?

Business intelligence refers to the technology tools and processes that enable businesses to organize, analyze, and contextualize business data from around the company. Business intelligence tools and decision-making transform raw data into meaningful and actionable information.

BI is the means through which organizations make smarter business decisions. While data fuels the engine, integrating BI-related infrastructure like a data warehouse, dashboards, reports, data discovery tools, and cloud data services make it possible to extract insights from your data.

THE ROLE OF BUSINESS INTELLIGENCE

Companies make big mistakes when they base business decisions on what they think will happen instead of relying on facts.

Using BI and advanced analytics, organizations can extract crucial facts from the mountain of data, transforming it into information companies can act on to make informed strategic decisions. The result: improved business processes, operational efficiency, and business productivity.

BUSINESS INTELLIGENCE DECISIONS

Business intelligence decisions typically fall into three categories: strategic, tactical, and operational.

An organization needs to gain a complete understanding of these types of decisions in business intelligence to make better-informed decisions that lead to increased customer retention, stakeholder satisfaction, operational efficiency, and revenue.

The Relationship between Business Intelligence and Business Analytics

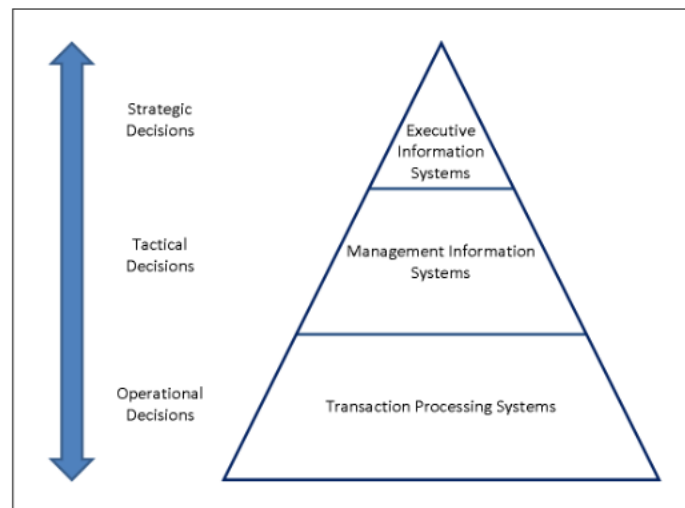
Business intelligence tells you what is currently happening and what happened in the past to bring you to that state.

On the other hand, business analytics is an umbrella term for predictive data analysis techniques (can tell you what's going to happen) and prescriptive (tells you what you should be doing to create better outcomes).

Using business intelligence and analytics efficiently is the difference between companies that succeed and those that fail in the modern environment.

THREE PRIMARY TYPES OF BUSINESS INTELLIGENCE DECISIONS

Business intelligence supports the three types of decision-making mentioned above: strategic, tactical, and operational. Its frequency and organizational impact characterize each.



STRATEGIC DECISIONS

Strategic decisions comprise the highest level of organizational business decisions and are usually less frequent and made by the organization's executives. Yet, their impact is enormous and far-reaching.

Some types of strategic decisions include selecting a particular market to penetrate, a company to acquire, or whether to hire additional staff.

Decisions made at this level usually involve significant expenditure. However, they are generally non-repetitive in nature and are taken only after careful analysis and evaluation of many alternatives.

TACTICAL DECISIONS

Tactical decisions (or semi structured decisions) occur with greater frequency (e.g., weekly or monthly) and fall into the mid-management level. Often, they relate to the implementation of strategic decisions.

Examples of tactical decisions include product price changes, work schedules, departmental reorganization, and similar activities. The impact of these types of decisions is medium regarding risk to the organization and impact on profitability.

OPERATIONAL DECISIONS

Operational decisions (or structured decisions) usually happen frequently (e.g., daily or hourly), relate to day-to-day operations of the enterprise, and have a lesser impact on the organization. Operational decisions determine the day-to-day profitability of the business, how effectively it retains customers, or how well it manages risk.

Answering a sales inquiry, approving a quotation, or calculating employee bonuses may be examples of this decision type.

You can summarize these types of decisions in business intelligence this way:

- Strategic: Long-term, complex, made by senior managers
- Tactical: Medium-term, less complex, made by mid-level managers
- Operational: Day-to-day, simple, routine, made by junior managers

HOW TO MAKE THE BEST DECISIONS FOR YOUR BUSINESS

How do you make the best business decisions? Some people trust intuition or gut feeling. Others reach out to constituents and experts for advice. Still, others cede decision-making to information systems and automation. However, the smartest business decisions are made by those who look at the numbers.

In a competitive business landscape, where agility, flexibility, and a real-time decision-making process are critical and timely, accurate data analysis is more important than ever. In that respect, relying on the types of decisions in business intelligence is non-negotiable. It is required for long-standing success and market dominance.

DECISION MAKING PROCESS**INTRODUCTION**

Decision making is the process of making choices by identifying a decision, gathering information, and assessing alternative resolutions.

Using a step-by-step decision-making process can help you make more deliberate, thoughtful decisions by organizing relevant information and defining alternatives. This approach increases the chances that you will choose the most satisfying alternative possible.

**Step 1: Identify the decision**

You realize that you need to make a decision. Try to clearly define the nature of the decision you must make. This first step is very important.

Step 2: Gather relevant information

Collect some pertinent information before you make your decision: what information is needed, the best sources of information, and how to get it. This step involves both internal and external “work.” Some information is internal: you’ll seek it through a process of self-assessment. Other information is external: you’ll find it online, in books, from other people, and from other sources.

Step 3: Identify the alternatives

As you collect information, you will probably identify several possible paths of action, or alternatives. You can also use your imagination and additional information to construct new alternatives. In this step, you will list all possible and desirable alternatives.

Step 4: Weigh the evidence

Draw on your information and emotions to imagine what it would be like if you carried out each of the alternatives to the end. Evaluate whether the need identified in Step 1 would be met or resolved through the use of each alternative. As you go through this difficult internal process, you'll begin to favor certain alternatives: those that seem to have a higher potential for reaching your goal. Finally, place the alternatives in a priority order, based upon your own value system.

Step 5: Choose among alternatives

Once you have weighed all the evidence, you are ready to select the alternative that seems to be best one for you. You may even choose a combination of alternatives. Your choice in Step 5 may very likely be the same or similar to the alternative you placed at the top of your list at the end of Step 4.

Step 6: Take action

You're now ready to take some positive action by beginning to implement the alternative you chose in Step 5.

Step 7: Review your decision & its consequences

In this final step, consider the results of your decision and evaluate whether or not it has resolved the need you identified in Step 1. If the decision has *not* met the identified need, you may want to repeat certain steps of the process to make a new decision. For example, you might want to gather more detailed or somewhat different information or explore additional alternatives.

CONCLUSION

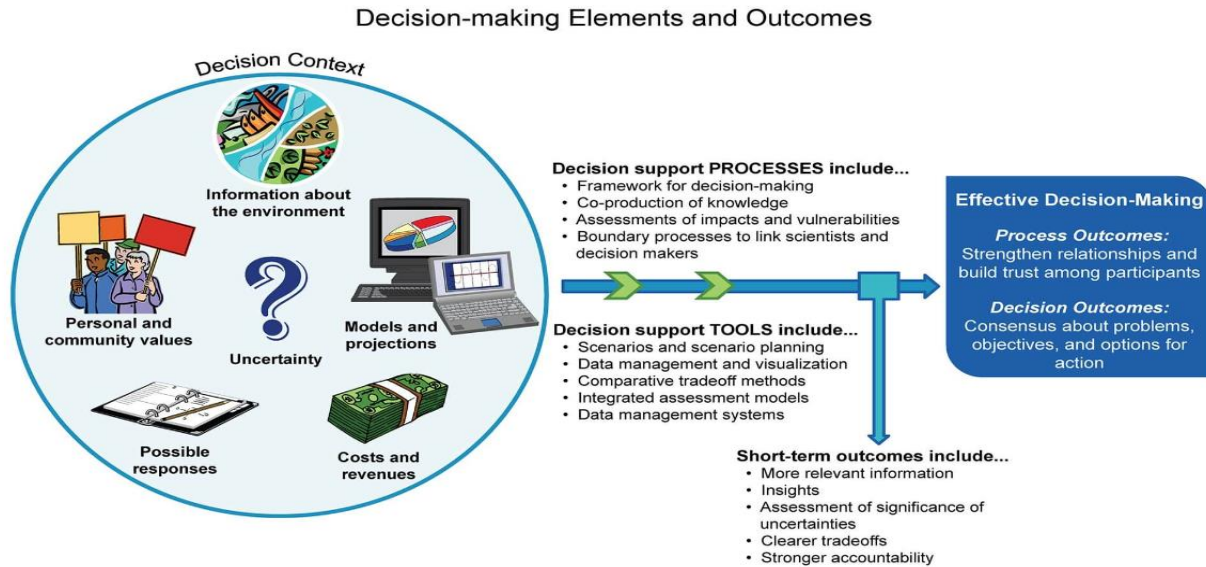
In any business situation there are multiple directions in which to take a strategy or an initiative. The variety of alternatives to weigh -- and the volume of decisions that must be made on an ongoing basis, especially in large organizations -- makes the implementation of an effective decision-making process a crucial element of managing successful business operations.

BUSINESS INTELLIGENCE IN DECISION SUPPORT SYSTEMS

A **decision support system (DSS)** is a computer-based information system that supports business or organizational decision-making activities; typically this results in ranking, sorting, or choosing from among alternatives. DSSs serve the management, operations, and planning levels of an organization (usually mid and higher management) and help people make decisions about problems that may be rapidly changing and not easily specified in advance.

There are several types of DSSs that include:

1. **Communication-driven DSS** which enables cooperation, supporting more than one person working on a shared task; examples include integrated tools like Google Docs or Microsoft Groove.
2. **Document-driven DSS** which manages, retrieves, and manipulates unstructured information in a variety of electronic formats.
3. **Knowledge-driven DSS** provides specialized problem-solving expertise stored as facts, rules, procedures, or in similar structures
4. **Model-driven DSS** emphasizes access to and manipulation of a statistical, financial, optimization, or simulation model. Model-driven DSS use data and parameters provided by users to assist decision makers in analyzing a situation; they are not necessarily data-intensive.
5. **Data-driven DSS** (or data-oriented DSS) emphasizes access to and manipulation of a time series of internal company data and, sometimes, external data. A data-driven DSS, which we will focus on, emphasizes access to and manipulation of a time series of internal company data and sometimes external data. Simple file systems accessed by query and retrieval tools provide the most elementary level of functionality. Data warehouse systems that allow the manipulation of data by computerized tools tailored to a specific task and setting or by more general tools and operators provide additional functionality. Data-driven DSS with online analytical processing (OLAP) provide the highest level of functionality.



What is a Decision Support System (DSS)?

A decision support system (DSS) is an information system that aids a business in decision-making activities that require judgment, determination, and a sequence of actions. The information system assists the mid- and high-level management of an organization by analyzing huge volumes of unstructured data and accumulating information that can help to solve problems and help in decision-making. A DSS is either human-powered, automated, or a combination of both.

Purpose of a Decision Support System

A decision support system produces detailed information reports by gathering and analyzing data. Hence, a DSS is different from a normal operations application, whose goal is to collect data and not analyze it.

In an organization, a DSS is used by the planning departments – such as the operations department – which collects data and creates a report that can be used by managers for decision-making. Mainly, a DSS is used in sales projection, for **inventory** and operations-related data, and to present information to customers in an easy-to-understand manner.

Theoretically, a DSS can be employed in various knowledge domains from an organization to forest management and the medical field. One of the main applications of a DSS in an organization is real-time reporting. It can be very helpful for organizations that take part in just-in-time (JIT) inventory management.

In a JIT inventory system, the organization requires real-time data of their inventory levels to place orders “just in time” to prevent delays in production and cause a negative domino effect.

Therefore, a DSS is more tailored to the individual or organization making the decision than a traditional system.

COMPONENTS OF A DECISION SUPPORT SYSTEM

The three main components of a DSS framework are:

1. Model Management System

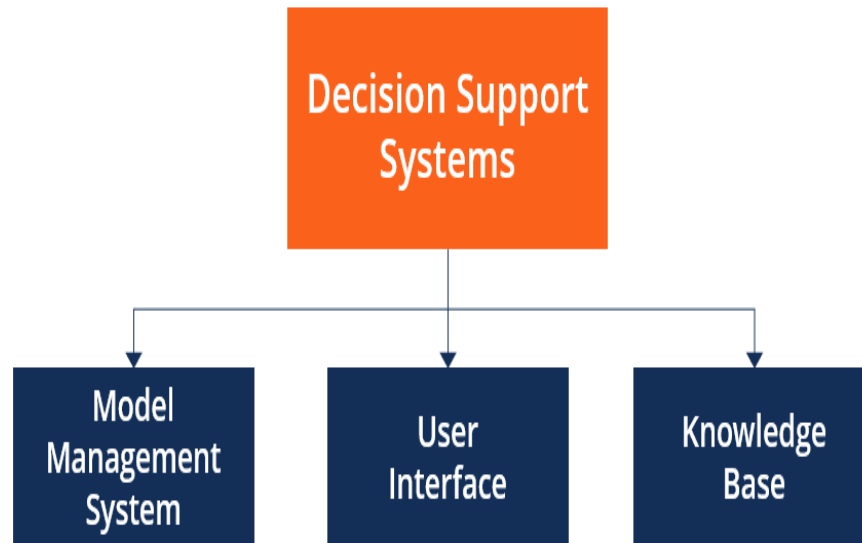
The model management system stores models that managers can use in their decision-making. The models are used in decision-making regarding the financial health of the organization and forecasting demand for a good or service.

2. User Interface

The user interface includes tools that help the end-user of a DSS to navigate through the system.

3. Knowledge Base

The knowledge base includes information from internal sources (information collected in a transaction process system) and external sources (newspapers and online databases).



Advantages of a Decision Support System

- A decision support system increases the speed and efficiency of decision-making activities. It is possible, as a DSS can collect and analyze real-time data.
- It promotes training within the organization, as specific skills must be developed to implement and run a DSS within an organization.

- It automates monotonous managerial processes, which means more of the manager's time can be spent on decision-making.
- It improves interpersonal communication within the organization.
-

Disadvantages of a Decision Support System

- The cost to develop and implement a DSS is a huge capital investment, which makes it less accessible to smaller organizations.
- A company can develop a dependence on a DSS, as it is integrated into daily decision-making processes to improve efficiency and speed. However, managers tend to rely on the system too much, which takes away the subjectivity aspect of decision-making.
- A DSS may lead to **information overload** because an information system tends to consider all aspects of a problem. It creates a dilemma for end-users, as they are left with multiple choices.
- Implementation of a DSS can cause fear and backlash from lower-level employees. Many of them are not comfortable with new technology and are afraid of losing their jobs to technology.

OLAP AND BUSINESS INTELLIGENCE

Business intelligence (BI) is an umbrella term for data analysis techniques, applications and practices used to support the decision-making processes in business. The term was proposed by Howard Dresner in 1989 and became widespread in the late 1990s.

Business intelligence assists business owners in making important decisions based on their business data. Rather than directly telling business owners what to do, business intelligence allows them to analyze the data they have to understand trends and get insights, thus scaffolding the decision-making process.

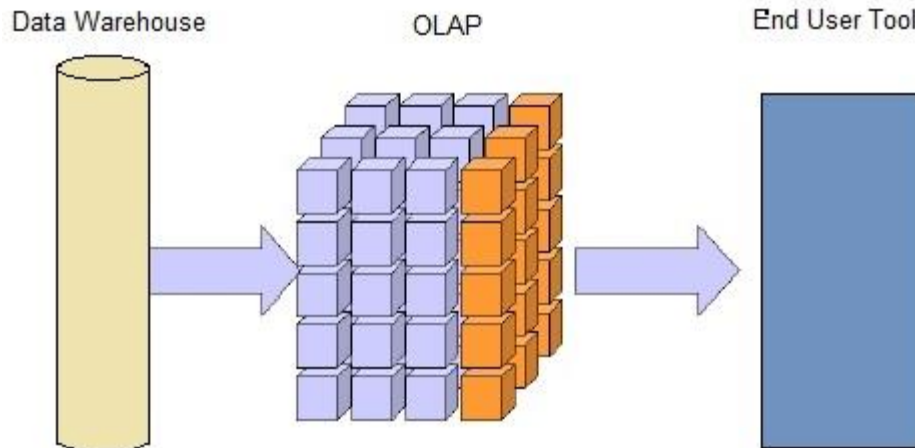
BI includes a wide variety of techniques and tools for data analytics, including tools for ad-hoc analytics and reporting, OLAP tools, real-time business intelligence, SaaS BI, etc. Another important area of BI is data visualization software, dashboards, and scorecards.

THE ROLE OF OLAP IN BUSINESS INTELLIGENCE

OLAP (online analytical processing) is sometimes used as a synonym of business intelligence. However, it is not correct - it could be better described a function of BI software that enables a user to extract and view data from different viewpoints.

There are several reasons why OLAP is popular in BI:

- It represents data in a multidimensional form, which makes it convenient for analysts and other business users to analyze numeric values from different perspectives.
- OLAP is good for storing, extracting and analyzing large amounts of data. Business intelligence specialists are able to analyze data accumulated over a long period of time, which enables more precise results and better forecasting. The architecture of OLAP systems allows fast access to the data as they typically pre-aggregate data.
- OLAP provides wide opportunities for data slicing and dicing, drill down/up/through, which helps analysts narrow down the data used for BI analysis and reporting.
- OLAP systems usually have an intuitive and easy-to-use interface, which allows nontechnical users to analyze data and generate reports without involving IT department. Also, OLAP dimension uses familiar business terms so that employees don't have to receive any additional training.



We can say that OLAP occupies a place between a data warehouse and end-user tools in BI, thus allowing users to get the data they need in a fast and efficient way.

What is OLAP?

OLAP (for *online analytical processing*) is software for performing multidimensional analysis at high speeds on large volumes of data from a [data warehouse](#), data mart, or some other unified, centralized data store.

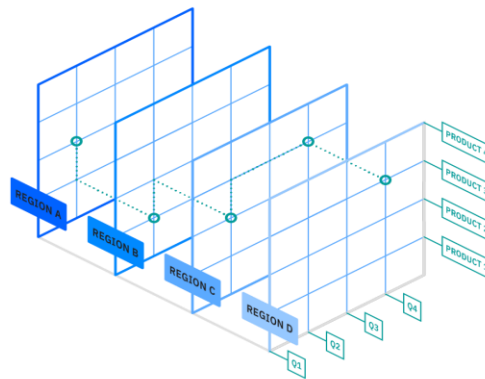
Most business data have multiple dimensions—multiple categories into which the data are broken down for presentation, tracking, or analysis. For example, sales figures might have several dimensions related to location (region, country, state/province, store), time (year, month, week, day), product (clothing, men/women/children, brand, type), and more.

But in a data warehouse, data sets are stored in tables, each of which can organize data into just two of these dimensions at a time. OLAP extracts data from multiple relational data sets and reorganizes it into a multidimensional format that enables very fast processing and very insightful analysis.

What is an OLAP cube?

This is where the OLAP cube comes in. The OLAP cube extends the single table with additional layers, each adding additional dimensions—usually the next level in the “concept hierarchy” of the dimension. For example, the top layer of the cube might organize sales by region; additional layers could be country, state/province, city and even specific store.

In theory, a cube can contain an infinite number of layers. (An OLAP cube representing more than three dimensions is sometimes called a hypercube.) And smaller cubes can exist within layers—for example, each store layer could contain cubes arranging sales by salesperson and product. In practice, data analysts will create OLAP cubes containing just the layers they need, for optimal analysis and performance.



OLAP cubes enable four basic types of multidimensional data analysis:

Drill-down

The drill-down operation converts less-detailed data into more-detailed data through one of two methods—moving down in the concept hierarchy or adding a new dimension to the cube. For example, if you view sales data for an organization’s calendar or fiscal quarter, you can drill-down to see sales for each month, moving down in the concept hierarchy of the “time” dimension.

Roll up

Roll up is the opposite of the drill-down function—it aggregates data on an OLAP cube by moving up in the concept hierarchy or by reducing the number of dimensions. For example, you could move up in the concept hierarchy of the “location” dimension by viewing each country’s data, rather than each city.

Slice and dice

The slice operation creates a sub-cube by selecting a single dimension from the main OLAP cube. For example, you can perform a slice by highlighting all data for the organization’s first fiscal or calendar quarter (time dimension).

The dice operation isolates a sub-cube by selecting several dimensions within the main OLAP cube. For example, you could perform a dice operation by highlighting all data by an organization’s calendar or fiscal quarters (time dimension) and within the U.S. and Canada (location dimension).

Pivot

The pivot function rotates the current cube view to display a new representation of the data—enabling dynamic multidimensional views of data. The OLAP pivot function is comparable to the pivot table feature in spreadsheet software, such as Microsoft Excel, but while pivot tables in

Excel can be challenging, OLAP pivots are relatively easier to use (less expertise is required) and have a faster response time and query performance.

Despite the variety and complexity of data stored in the corporate environment, everything is typically recorded in simple columns and rows. This is the classic spreadsheet look we're all familiar with, and that's how most databases file data.



An example of database tables, structuring music by artists, albums, and ratings dimensions

For the most part, businesses use databases to record transactions. This is an operational need, as we have to save our sales results, customer information, etc. Later, this data can be

- modified to maintain the relevance of what was stored;
- used by business applications to perform its functions, for example check product availability, etc.; and
- used for analytical purposes to understand how our business is running.

The database type that stores transactions is called *OLTP* or *Online Transaction Processing*. It perfectly suited for simple operations like to insert, update, or delete an item. However, when it comes to analyzing large volumes of data from different angles, the logic of OLTP has serious limitations. So, we need a solution that's capable of representing data from multiple dimensions.

In this article, we'll talk about such a solution — *Online Analytical Processing* or *OLAP* technology. We'll cover the fundamentals of OLAP and see how it works in contrast to transactional databases. Namely, we'll explain what functions it can perform, and how to use it for data analysis.

What is OLAP: Online Analytical Processing

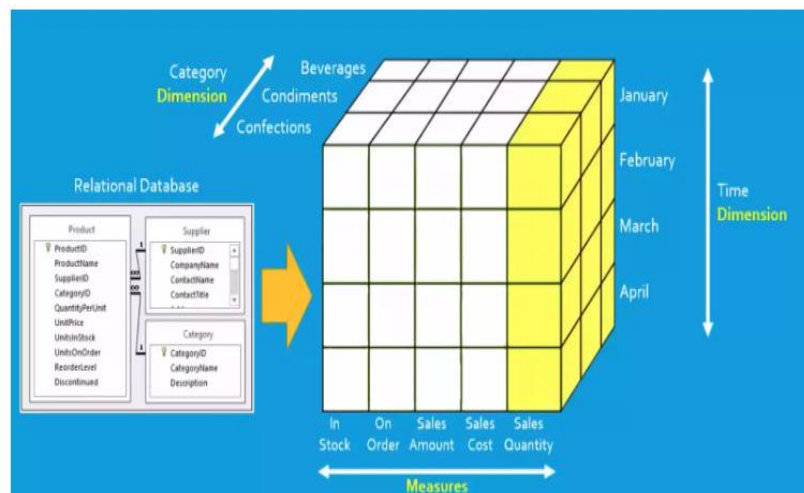
Online Analytical Processing can be defined as a set of tools and approaches to represent data from multiple dimensions. In a broader sense, it includes a bunch of practices aimed at modeling data/databases and creating specific analytical solutions. OLAP systems are capable of combining classic tables in a sort of table of tables, which can be visualized as a 3D OLAP Cube for simplicity.

A typical OLAP system will include the following components that perform dedicated functions to handle analytical queries.

Data source. This could be a transactional database or any other storage we take data from. The data in its standard format isn't optimized for OLAP queries, so it requires transformation and remodeling before it can be used.

OLAP database is where we store data for analysis. Usually, transformation takes place before the data is uploaded to a database, but the approach may vary.

OLAP cube is basically a tool for representing multidimensional data for analysis. As we're talking about online analytical processing, cubes are deployed on a dedicated server.



OLAP Cube representing data from an OLTP database in multiple dimensions

An OLAP cube allows analytics to group or slice items by different categories. They are primarily designed to run complex queries, which can't be handled by the usual OLTP databases. Here, a user can perform cube-specific operations with data, so we'll cover them in a dedicated section.

Analytical interface- The interactions with cubes and other analytical tools for [data visualization](#) and reporting is done via a dedicated interface. The majority of interfaces are represented by business intelligence dashboards. Cubes can be accessed via these dashboards, providing more control to a user. The easiest way to understand how OLAP works is to compare OLAP and OLTP databases and explore how they structure and process data.

OLTP VS OLAP: TECHNOLOGY COMPARISON

There are numerous differences between OLTP and OLAP databases in terms of purpose, information structure, and data access capabilities. The table below compares the main aspects of these two systems.

OLTP VS OLAP		
DB Type	OLTP Database	OLAP Database
Purpose	- Collect and store transactional data. - Maintain data integrity. - Process queries to support business processes run by applications or employees.	- Aggregate transactional data for analysis. - Support business decision making. - Discover trends and insights.
Query type	Simple queries to run commands like: INSERT, UPDATE, DELETE	Complex queries with custom commands
Data source	Transactions	Aggregated transaction data
Data update	Fast updates on separate data points, or small batches	Large, or usually full batch updates
Data view	Flat two-dimensional view	Multidimensional view
Transaction duration	Short transaction (response measured in milliseconds)	Long transactions (response measured in minutes or hours)
User	Operational staff or business applications	Data analysts, business analysts, managers of all levels

Now, we'll look in more detail at how both types can be used, what operations they run, and how the data is structured for OLTP and OLAP purposes.

Data operations in OLTP

A transactional or OLTP database is a common storage solution we deal with to record any business information. Say, we're selling a new type of a smartphone to the customer and we want to record this transaction, including the product type, price, date, customer info, sales person name, etc. All of these items will be stored in a flat view, which allows us to quickly operate and search for the required information.

The data will be saved as a set of items and values that relate to this transaction. An OLTP solution will allow a user to perform the following operations with this data:

- insert,
- copy,

- paste,
- edit/update, and
- delete.

Such transactions have a short response time – measured in seconds – as they are natural to OLTP. But when it comes to more complex queries that involve aggregating data from multiple tables, a transactional database will run into trouble. The more data is inquired, the more problematic and resource-intensive it is for OLTP.

Analytical requests are often much more complex than “show me total sales amount.” More often than not, we need to compare things to each other and look at the data from different dimensions. That’s where an OLAP technology kicks in.

Data operations in OLAP

OLAP or Online Analytical Processing aggregates transactional data from a storage to transform it into a feasible form for analysis. As a source of data, OLAP can use some type of unified storage like a data warehouse, [data lake](#), or [data mart](#), or simply any place you store the historical data.

But to run complex custom queries, we must structure data properly. That’s why in most cases, there is a need for a separate OLAP database or warehouse that will model data for multidimensional analysis.

The query in OLAP may look something like this:

- *“show 64GB smartphone sales in Winnipeg for the last 6 month,”*
- *“compare 64GB model sales across Canada to 256GB in the 4thQ,”*
- *“group all 64GB model returns for the 2021 sold by John Doe salesperson,”* and
- *“show average margin for a given smartphone vendor for this year.”*

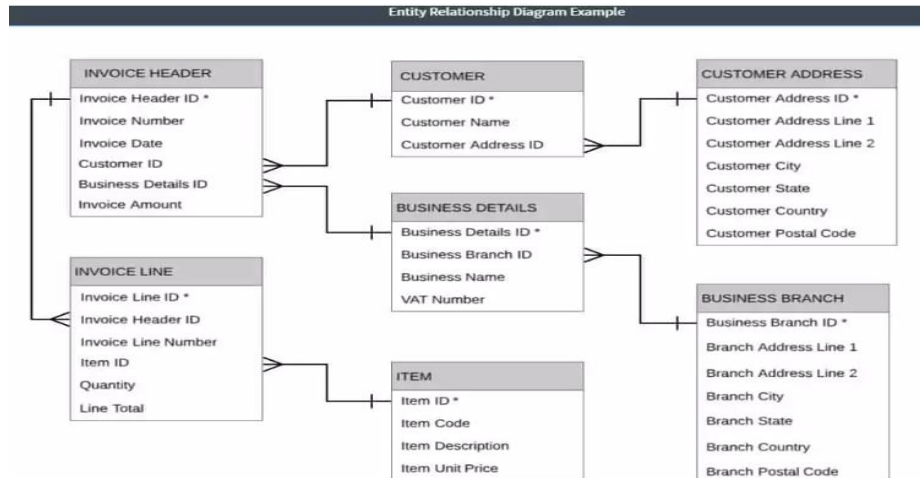
Such analytical queries require the database to gather information from multiple tables that categorize data by “dimensions.” An example of dimension can be time, product, location, customer, etc.

OLAP models a database in such a way that it becomes possible to quickly gather the data and present it to analysts in a multidimensional mode rather than a flat table. That’s why OLTP and OLAP databases will differ in numerous ways.

Now, we have to answer two simple questions. How is OLAP data modeling different from transactional databases? And why can’t we run such complex queries in OLTP?

Data modeling in OLTP

Traditional databases (OLTP) use a **relational data model**, hence the name “relational database.” The relations are nothing but tables of values. Each row in the table denotes a real-world relation of an item to its attribute. For example, a customer may have related attributes such as address, email, credit card, name, etc.



Relational data model example

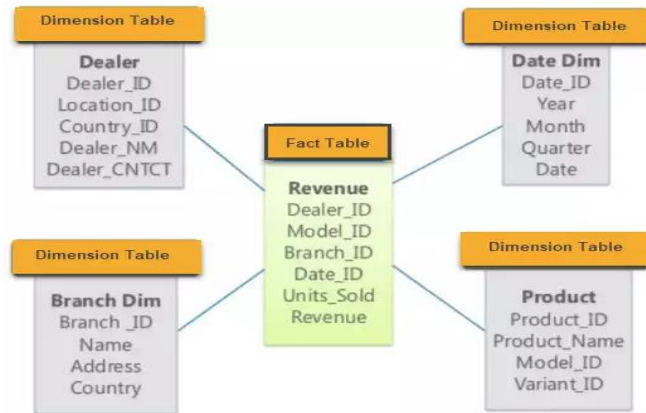
This is the most standard way we store data and make modifications to transactional information. Such an approach works great for simple queries to modify transactional data. But if we need to query something like “compare sales of a given item in the 3rd quarter for the last three years in the US” — the relational database will require enormous resources because it will scan each table entirely to find all the related values.

Moreover, the query will return disparate data items with a lot of unnecessary information, as the relational model doesn’t support filtering by multiple dimensions at once (product type, time period, location).

Data modeling in OLAP

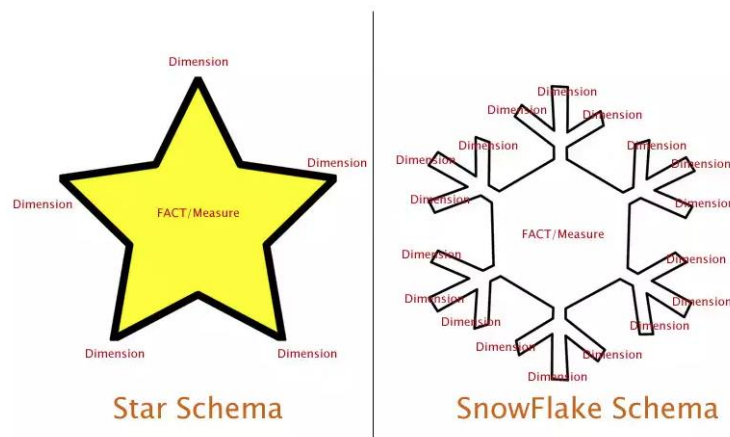
In contrast, OLAP uses *star* and *snowflake schemas* for data modeling.

In a **star schema** we structure data around facts, providing the keys to every dimension for measurement. A fact, in this case, is a category of related business items, e.g., product, sales amount, revenue, customers, time, location, etc. Each of these items is a separate dimension that includes subcategories. So we can divide, for example, time by year, quarter, month, week, and day.



Star schema in OLAP

A **snowflake schema** is an extension to a star schema: Basically, it adds more dimensions to the existing ones. But data is still organized around factual tables. So if we add more dimension tables, the database starts to take a “snowflake shape.”



A snowflake schema extends the star schema with more dimensions

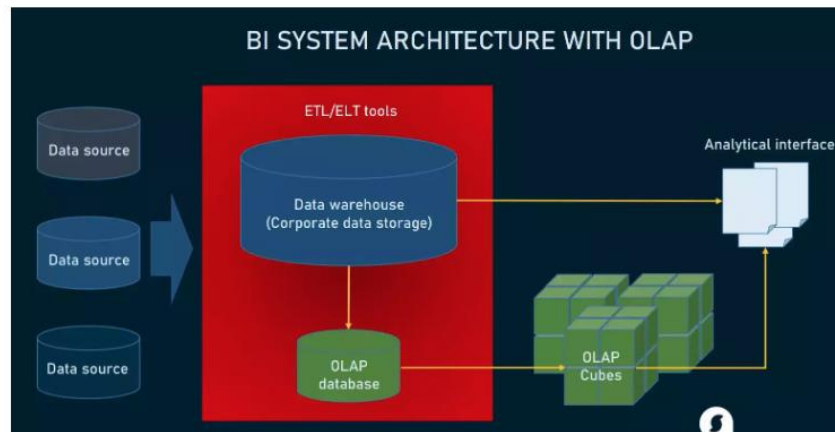
A multidimensional model of data is what makes it possible for OLAP systems to extract the required information, perform complex filtering, and allow for analysis of this data.

Online Analytical Processing Architecture

When we talk about analytics, we generally mean business intelligence systems, where a data warehouse is a central point uniting all the corporate data. A data warehouse can be used differently depending on the goals of the organization.

- A corporate DW can be used as a unified storage for historical data, optimized for transaction or simple analytical tasks.
- Or it can be optimized for OLAP queries applying star or snowflake schemas.

Depending on how we plan to [design the business intelligence](#) system, OLAP may or may not require a separate database to run queries. The architecture of a BI system with a standalone OLAP repository looks something like this:



So let's analyze OLAP workflow in such architecture.

Data extraction. First, the data is extracted from its original sources and uploaded to a unified data storage. In the case of BI, a data warehouse will be the place we upload data to.

Data preparation. Once we've got the data, it requires optimization and modeling for multidimensional analysis. In some cases, corporate DW can be optimized to run OLAP queries, but a more typical case is to use a separate OLAP database. Here are a few reasons why.

- Running analytical and transactional queries on separate databases eliminates the risk of overloads and database downtimes, while guaranteeing decent performance of the two.
- Applying data models is easier when we use a storage for a single purpose.

Data transformation and integration is usually done via [ETL/ELT](#) tools, which help developers to automate data extraction, transformation, and uploading.

Building a cube. Once the data is prepared, a group of responsible data engineers will model cubes and deploy them on the dedicated server. Creating a cube is a custom process each time, because data can't be updated once it was modeled in a cube. So, for each specific query, a new cube will be created.

Accessing data. As an end point in the system, OLAP cubes will be accessed through analytical interfaces. Here, analytics can type in commands and perform cube-specific operations to analyze data.

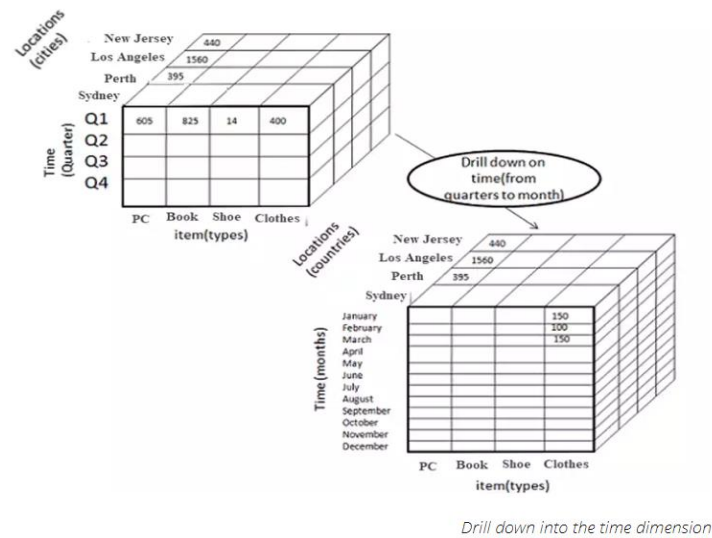
Now, let's look at the cubes themselves and define the capabilities they give to the analyst.

How OLAP cubes work

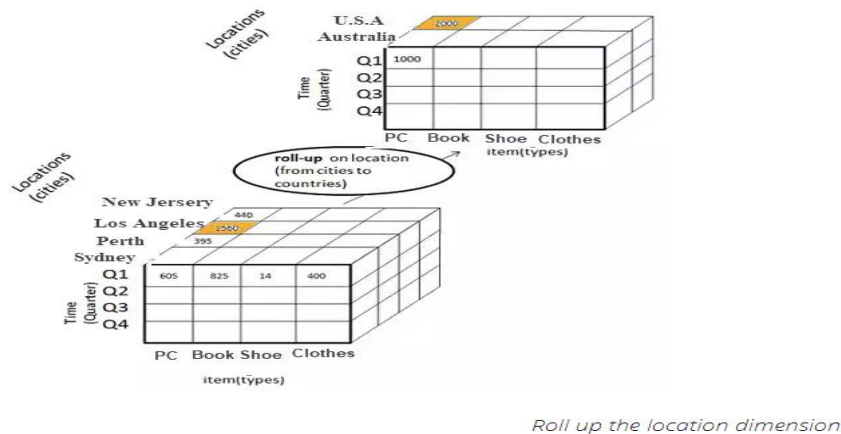
The cube may consist of several dimensions that can be used to filter the required information and form reports. OLAP systems use a specific SQL language called *MDX* or *Multidimensional Expressions*. Standard SQL queries are also supported by the most databases to perform OLAP analysis.

There are five basic types of OLAP multidimensional analysis operations.

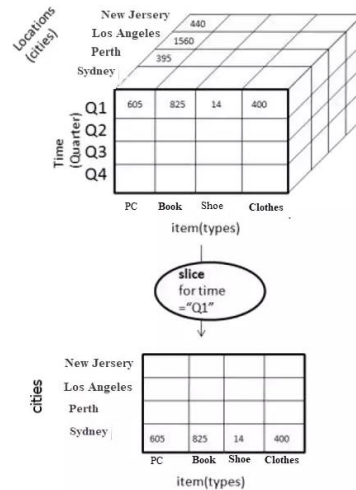
Drill down allows a user to move from high-level data (e.g., annual sales) to a lower level (e.g., monthly sales). Here we use the concept of hierarchy that applies to every single dimension. So, in the “time” dimension, we can move down from yearly figures to weekly or even daily records. This depends on how you store your data and model the actual cube.



Roll up is the opposite of drill down, as it basically lifts the data in hierarchy levels. Both operations either make the data more or less detailed, or add/remove dimensions for the analysis.

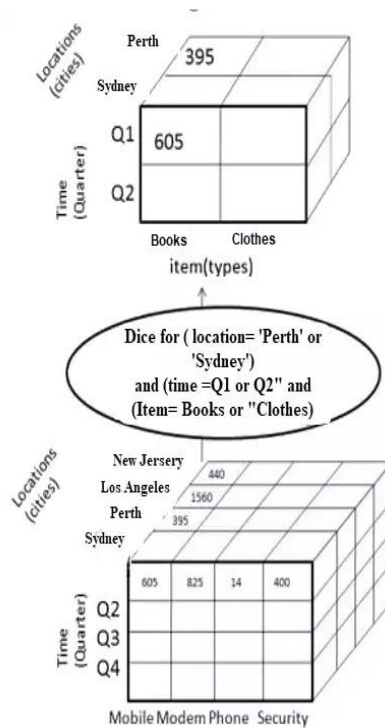


Slice operations help you divide a certain dimension into a separate table (one-dimension view). “Slice” can detach, say, the city’s dimension from the rest of the cube, which will create a separate spreadsheet. This way we can analyze low-level information in the isolated environment.



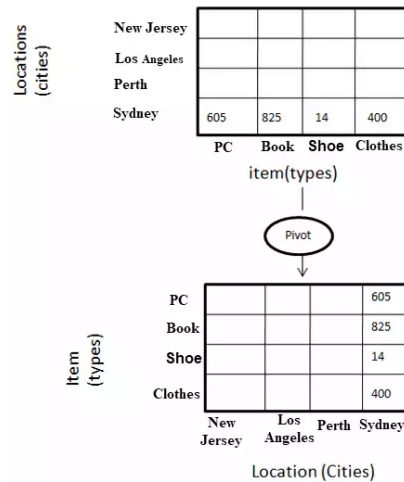
Slicing a location dimension in a hierarchy level of cities

Dice provides the same separation functionality, but allows you to choose more than one dimension, producing a separate cube.



A dice operation with the choice of time and location dimensions

Pivot is a similar operation to create pivot tables in Excel. This function allows us to rotate a cube to get a different representation of data in between the dimensions.



Pivot example on a single dimension of the cube

All in all, the functions can be used in conjunction, which gives huge flexibility to use a single cube for multiple purposes. But, as we mentioned before, each time there is a modification to data, the cube will require reuploading the information or remodeling the existing OLAP DB.

OLAP PROVIDERS

OLAP is a vital part of any BI system. Despite its resource-intensive nature, OLAP remains a standard solution for complex analytics that can't be done in the usual databases. As the technology appeared in the early '90s, the market of solutions is quite large. And the main proposal comes from data warehouse/business intelligence providers.

Nearly any provider these days supports all of the basic functions of OLAP and allows the creation of multidimensional cube systems as a part of their BI platform. Now, let's look at some popular products that can be used as a separate OLAP tool.

OLAP PROVIDERS			
	Apache Kylin	IBM Cognos TM1	Microsoft SSAS
Supported OS	Linux/Unix	Windows, Linux/Unix	Windows
Supported APIs	JSON, REST APIs	JSON, REST APIs	JSON, REST APIs
Relational database support	NO	NO	YES
Offline access to cubes	YES	YES	YES
Supported Query language	SQL	XML, MDX, LINQ	XML, MDX, SQL, LINQ
Cube limitations	Unrestricted to dimension and cubes number	256 dimensions max limit for a single cube	Unrestricted to dimension and cubes number
License pricing	Free	By inquiry	Calculated from an hourly/monthly data transmission

[Apache Kylin](#) is an open-source distributed data warehouse for big data and OLAP. Kylin was developed for internal analytics on Ebay. Since 2014, it has gone open source and is distributed by a free license. While it focuses on analyzing big data, Kylin can also be used for corporate warehouses of medium size. Plus, Kylin integrates with the popular BI interfaces such as [Tableau](#), Superset, Qlik, and Zeppelin.

[Microsoft SQL Server Analysis Services \(SSAS\)](#). As a part of its Azure Cloud Platform and PowerBI analytical solutions, Microsoft offers a separate product for OLAP. Currently, they call it Azure Analysis Services. Basically, it's an OLAP modeling and processing tool integrated with PowerBI. The pricing is calculated like all the Azure products, based on computation resources. You can check [pricing](#) on the corresponding page.

[IBM Cognos TM1](#) is another platform that consists of multiple tools for data analysis, cube modeling, and data visualization. As with Microsoft, TM1 includes a broad range of Watson Analytics products and a dedicated analytical server. The pricing for TM1 can be gotten from IBM on their [price planning](#) page.

But while software vendors offer tools for data modeling and analysis, you'll still need data engineers/data analysts to model and analyze the information. In other words, OLAP falls into a category of specific analytical tools that require [data engineering](#) or [BI development](#) expertise to work with. In this case, you may think of hiring a [data specialists team](#) to handle a custom project for your corporate needs.

THE RISE OF COLUMN DATABASES OR A NEW OLAP

Currently, OLAP remains quite a cumbersome technology, because it requires modeling a separate database to build cubes. And the more data you need for analysis, the more likely you'll need a data warehouse only for OLAP needs. But things may change with the emergence of **column databases**.

Row-oriented			
ID	Name	Grade	GPA
001	John	Senior	4.00
002	Karen	Freshman	3.67
003	Bill	Junior	3.33

Column-oriented					
Name	ID	Grade	ID	GPA	ID
John	001	Senior	001	4.00	001
Karen	002	Freshman	002	3.67	002
Bill	003	Junior	003	3.33	003

A comparison of a relational DB row-table, and a column-table.

As you might remember, a traditional relational database stores values in rows, while columns denote categories of items. A column database is a type of schema that uses columns to organize tables in DB. As simple as that, this type of schema provides capabilities similar to what an OLAP database does. Each table will represent a dimension that can be quickly scanned and analyzed. Column databases can potentially be used as a data warehouse capable of handling OLAP queries by nature. While this approach was described way back in 2012 in different [studies](#), it gained popularity only a few years ago. So this led to the emergence of column-oriented [cloud data warehouses](#).

- [AWS Redshift](#) offers columnar storage as an option for high-volume analytical queries. The concept of a column database in Redshift is described on a [dedicated page](#).
- [Apache Kudu](#) is an analytical data store that relates to the [Apache Hadoop](#). Kudu is an open-source solution designed specifically for analyzing dynamically changing data.
- [MariaDB ColumnStore](#) is a distributed data storage solution based on the [MariaDB](#) Enterprise server.
- [Google Cloud Bigtable](#) is a part of the Google Cloud ecosystem designed for large volume data analysis, big data, and [streaming analytics](#).
- [Azure Synapse Analytics](#) connects a range of services for data warehousing, [data integration](#) (ETL), and big data analytics.

Choosing a column-oriented database eliminates the need to create a separate ecosystem for OLAP. But there is a drawback to keep in mind. In case you'll have to run a massive update of your data, a column database will need to read every single row one by one. The process of updating currently remains much more time-consuming and complex for column databases than it is for traditional SQL databases. What does it mean in the context of analytics? Historical data often doesn't require any updates, so if you're analyzing large amounts of transactional data from the past, the updating issue is not really valid. But in case you're trying to analyze dynamically updating data, a traditional SQL database remains the best choice.